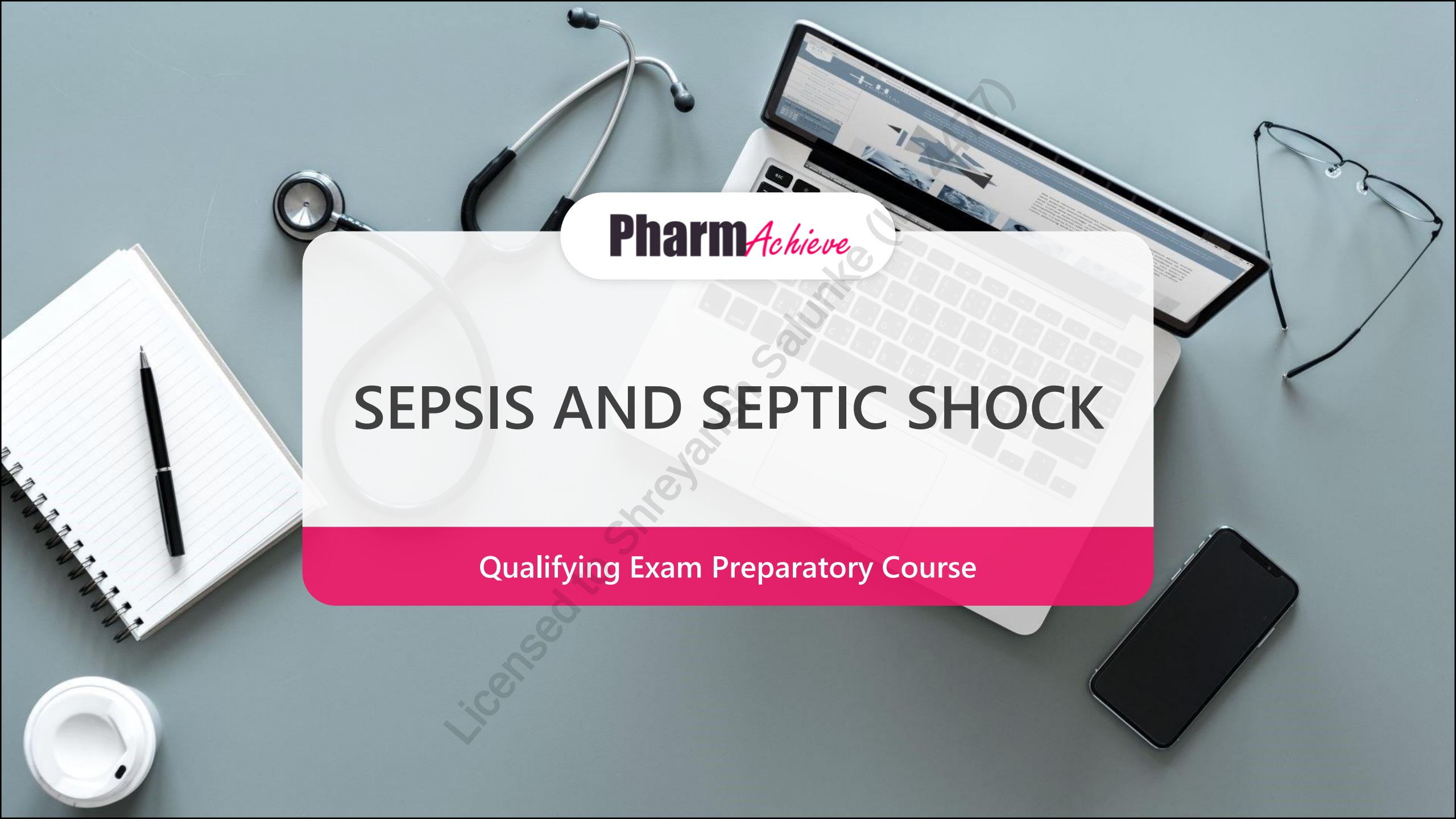




Pharm*Achieve*

SEPSIS AND SEPTIC SHOCK

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AVPU scale	Alert, Voice, Pain, Unresponsive
BP	Blood Pressure
Bpm	Beats Per Minute
CVC	Central Venous Catheter
CVP	Central Venous Pressure
FiO2	Fraction of Inspired Oxygen
HR	Heart Rate
IVC	Inferior Vena Cava
MAP	Mean Arterial Pressure
POCUS	Point-of-Care Ultrasound
RR	Respiratory Rate
SBP	Systolic Blood Pressure
SIRS	Systemic Inflammatory Response Syndrome
ScvO₂	Central Venous Oxygen Saturation

WHAT IS SEPSIS?

- Sepsis is a **dysregulated host response to infection**
- Sepsis is **life-threatening**, therefore early detection and prompt management are crucial
- Sepsis can progress to **septic shock**
- Various validated tools exist to screen for sepsis. These should be used in combination with clinical judgement
- **SIRS is typically not used anymore; more common screening tools are NEWS, MEWS, qSOFA**

SCREENING TOOLS

FYI

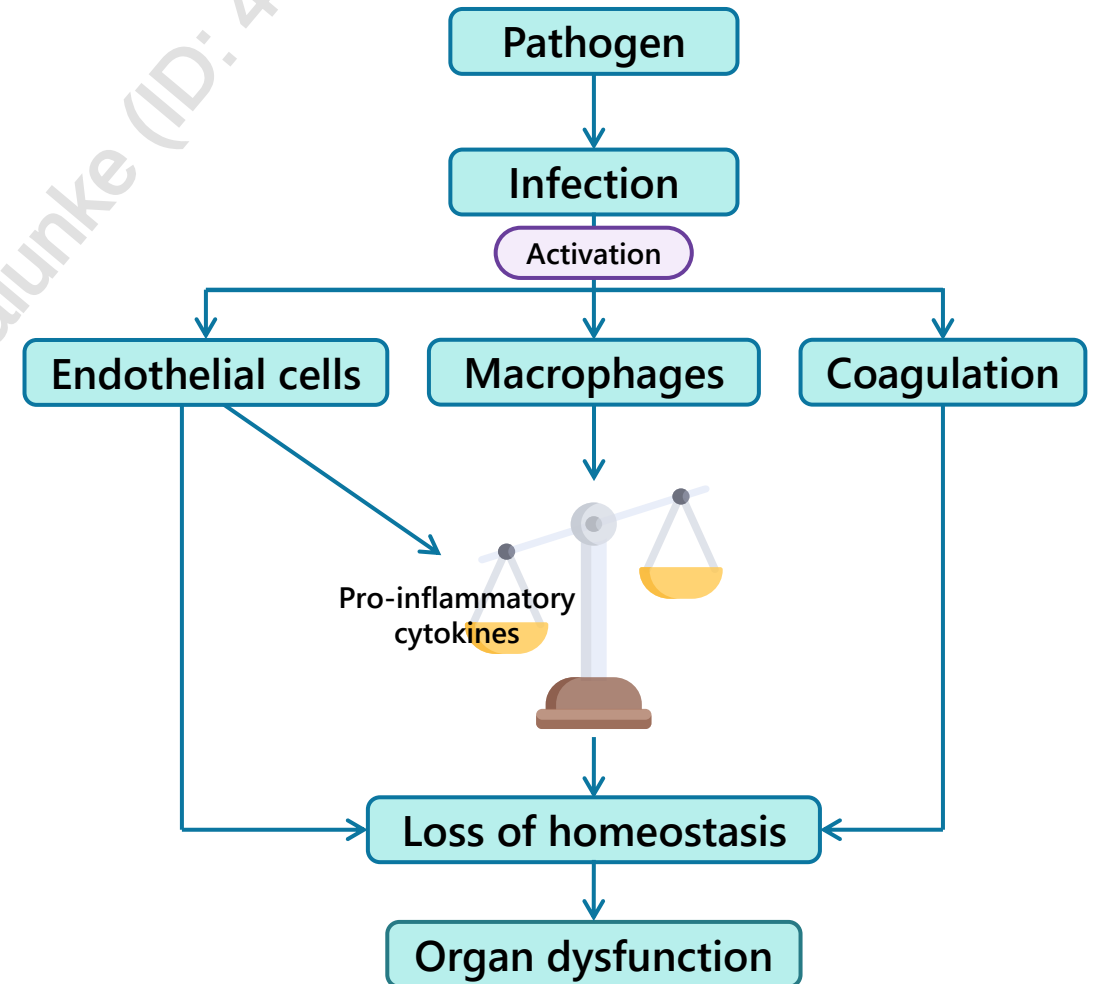
National Early Warning Score (NEWS)	Modified Early Warning Score (MEWS)	Sequential organ failure assessment (qSOFA)
Respiratory rate	Respiratory rate	Respiratory rate ≥ 22 /minute
Systolic blood pressure	Systolic blood pressure	Systolic blood pressure ≤ 100 mmHg
Heart rate	Heart rate	-----
Level of consciousness (AVPU scale)	Level of consciousness (AVPU scale)	Level of consciousness (Glasgow Coma Scale)
Temperature	Temperature	-----
Need for supplemental oxygen		-----
Oxygen saturation		-----
Score: 0-3 for each parameter	Score: 0-3 for each parameter	Positive score: $\geq 2/3$ criteria met

Identifies patients who are at an increased risk of morbidity and mortality

PATHOPHYSIOLOGY

FYI

- Body creates a **localized inflammatory response** to an invading pathogen
- Inflammatory response is uncontrolled and moves **beyond the localized area**
 - Cytokine production
 - T-cell recruitment and activation
 - Bone marrow stimulation and leukocytosis
 - Macrophage demargination
- Generalized response:
 - Cellular injury
 - Organ dysfunction



RISK FACTORS

- Age (very young, older $\geq$ 65 years)
- Genetics
- Prolonged hospital stay and/or previous hospitalization
- Bacteremia/systemic infection
- Comorbidities (diabetes, obesity, heart failure, cancer, indwelling catheter)
- Immunocompromised status

GOALS OF THERAPY

- ✓ Timely diagnosis and treatment
- ✓ Initiation of antimicrobial therapy
- ✓ Identify causative pathogen
- ✓ Prevent complications

THERAPEUTIC APPROACH

Stabilize breathing

- Supplemental oxygen, intubation, mechanical ventilation



Establish venous access (central venous catheter)

- Drug administration, may need to place CVC, monitor CVP and ScvO2



Organ and tissue perfusion

- Assess for hypotension and hypoperfusion
- Aggressive administration of IV fluids started within 1 h of presentation and completed within next 3 hours
- Goal within first 6 hours: CVP 8-12 mmHg, MAP >65 mmHg, urine output >0.5 mL/kg/hour, ScvO2 >70% or mixed venous oxygen saturation >65%



Antimicrobial therapy

- Timely (within first hour) administration of antibiotics decreases the risk of mortality



Memory Tool:

BAPA

1. Breathing
2. Access
3. Perfusion
4. Antibiotics

THERAPY FOR ORGAN AND TISSUE PERFUSION

IV fluids

- **Crystalloids** (e.g. normal saline, Ringer's lactate): treatment of choice to restore perfusion for severe sepsis/septic shock; can be continued as long as the patient is improving hemodynamically
- **Colloids** (e.g., albumin solutions): non-inferior to crystalloids, more expensive
- Side effects of both: volume overload, pulmonary edema, prolonged ventilation
- No advantage to dextran solutions
- Hyperoncotic starch is not recommended due to increased risk of kidney dysfunction and mortality

DRUG THERAPY FOR ORGAN AND TISSUE PERFUSION

Fluid resuscitation: vasopressors

- Use vasopressors in patients unresponsive to adequate fluid resuscitation and to restore target MAP and reduce signs of hypoperfusion
- Norepinephrine (first-line)
 - Potent vasoconstrictor ($\alpha_1 \gg \beta_1 \gg \beta_2$); less inotropic and chronotropic effects compared to epinephrine and dopamine
- Epinephrine:
 - May replace/ be added to norepinephrine to reach target MAP ($\beta_1 \gg \beta_2 = \alpha_1$); positive inotrope
- Dopamine
 - Acts on dopamine-1, α_1 and β_1 ; affects both vascular resistance and cardiac output
 - Increased risk for atrial arrhythmia and mortality due to β_1 effects
 - Use is limited by its strong positive chronotropic effects (increased heart rate)
- Phenylephrine: effective in patients with tachycardia or arrhythmias (α_1 only)
- Vasopressin: adjunctive therapy (NOT monotherapy) to norepinephrine to minimize use of high-dose norepinephrine

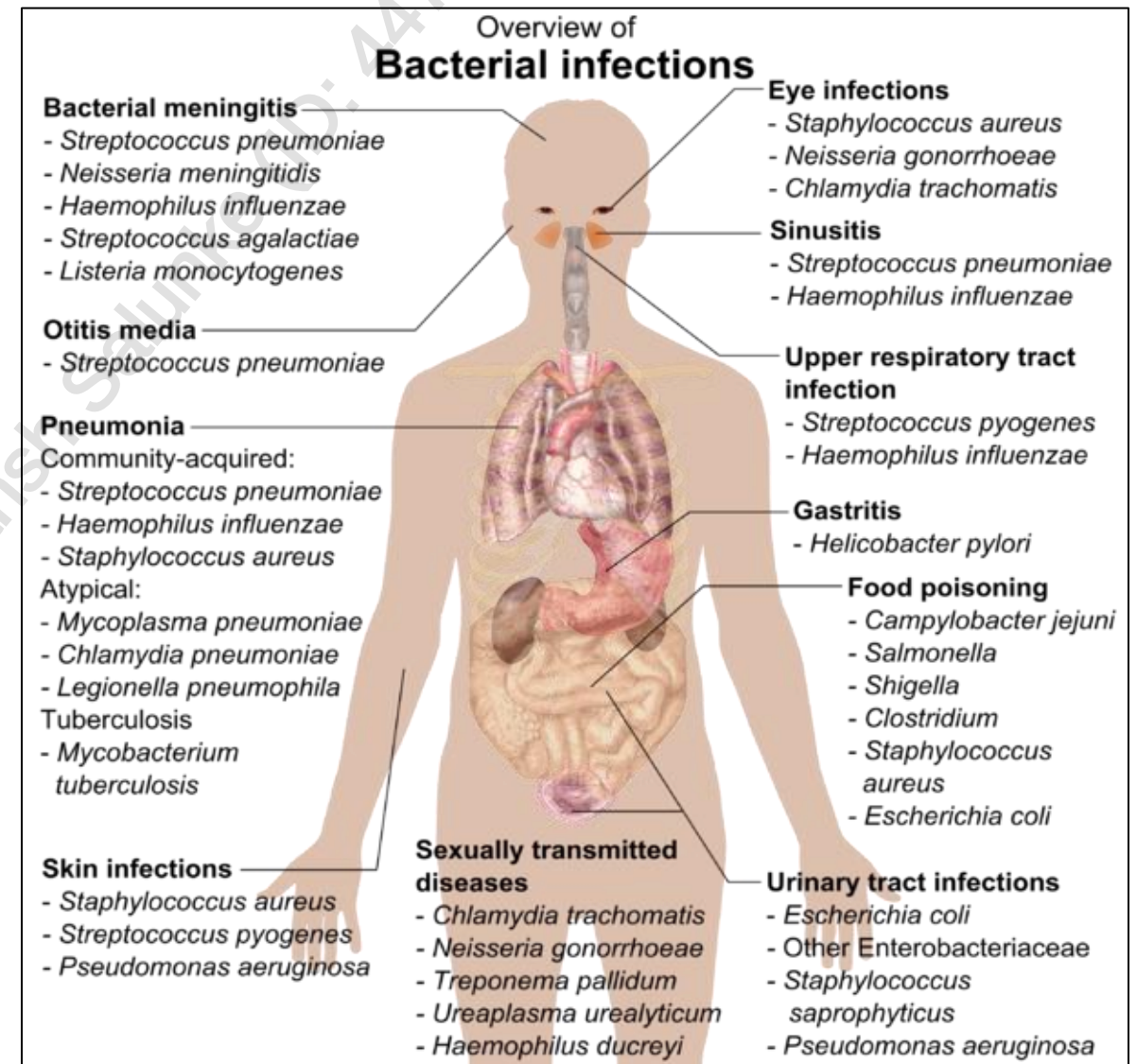
DRUG THERAPY FOR ORGAN AND TISSUE PERFUSION

Fluid resuscitation: other therapies

- **Dobutamine**
 - Mainly β_1 , but some α_1 and β_2 activity causes vasodilation which may reduce MAP
 - **Increases cardiac output and improves oxygen delivery**
 - Useful in late shock or in patients with HF as it is a positive inotrope
- **Blood product transfusions**
 - Use blood **only in patients with hemoglobin concentration <70 g/L or evidence of active hemorrhage;**
- **Corticosteroids**
 - Guidelines recommend against routine use of glucocorticoids (does not improve mortality) thus **only use as adjunctive therapy when required**
 - Some studies suggest an increased risk of superinfections and new septic shock with the use of corticosteroids
 - Suggested in those with pre-sepsis steroid usage or Addison's disease
 - 200 mg/day IV hydrocortisone in divided

ANTIMICROBIAL THERAPY

- Administer **broad-spectrum IV antibiotics as soon as possible**
 - Mortality increases dramatically with each hour that antibiotic therapy is delayed
- **Initial therapy is empiric**; consider site of infection, patient history, local resistance patterns
- **Obtain cultures to choose more narrow spectrum antibiotics**
- **Reassess patient daily**: drain abscesses, monitor clinical response and toxicity
- **Duration depends on source of infection, microbe, host-factors, response**



INTRAVENOUS ANTIMICROBIAL THERAPY

Broad Spectrum Antibiotics	Others	Antifungal
Penicillin w/ beta-lactamases • Piperacillin-Tazobactam • Ticarcillin/Clavulin	• Daptomycin • Vancomycin • Linezolid • Tigecycline • Metronidazole • Macrolides Aminoglycosides • Amikacin • Gentamicin • Tobramycin	Azole • Fluconazole • Voriconazole
Cephalosporin (Extended Spectrum) • Cefotaxime (3rd gen) • Ceftazidime (3rd gen) • Ceftriaxone (3rd gen) • Cefepime (4th gen)		Echinocandin • Anidulafungin • Caspofungin • Micafungin
Carbapenem • Ertapenem • Imipenem-cilastatin • Meropenem		Polyene Antifungals: • Amphotericin B
Fluoroquinolones • Ciprofloxacin • Levofloxacin • Moxifloxacin		

PREVENTATIVE MEASURES WHILE IN ICU

- **Glucose control:** IV insulin
- **Venous thromboembolism:** Unfractionated heparin, low molecular weight heparin
- **Stress ulcer:** H₂ receptor blockers
- **Ventilator-associated pneumonia:** mouth care, head of bed at 30 degrees
- **CNS dysfunction:** discontinue precipitating agents, administer dexmedetomidine as a sedative or treatment of ICU delirium. Antipsychotics to manage delirium (should be last option, only if failed other measures),
- **Nosocomial infections:** handwashing, aseptic technique and other infection control measures
- **Adequate caloric intake:** vitamins, minerals, trace elements through enteral route
- **Impaired GI motility:** use prokinetic agents such as metoclopramide or erythromycin
- **Edema:** diuretics (don't use with vasopressors)

TAKEAWAYS

Therapeutic Approach - **BAPA**:

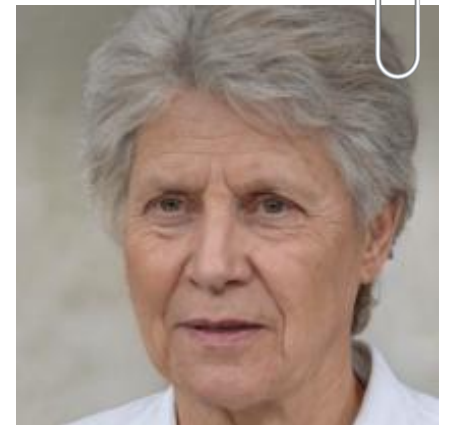
- **B**reathing (stabilize with oxygen, intubation, or mechanical ventilation)
- **A**ccess (acquire venous access – monitors CVP and allows drug administration)
- **P**erfusion (IV fluids and possible vasopressors to target CVP 8-12 mmHg, MAP > 65 mmHg, and ScvO₂ > 65% oxygen saturation).
- **A**ntibiotics (acquire samples for culture and start empirically within 1st hour)

Antimicrobials

- Choice based on source of original infection
- Broad spectrum agents +/- other agents for specific risk factors (PsA, VRE, MRSA) +/- anti-fungals
- Narrow according to cultures and susceptibility

CASE 1

CR is an 80-year-old female who lives in a long-term care facility and is becoming increasingly lethargic. Four days ago, she was placed on nitrofurantoin macrocrystals 100 mg PO BID x 7 days for a presumed urinary tract infection. The nursing staff noticed she was becoming increasingly drowsy. Today, CR is barely responsive. CR has a past medical history of hypertension, dementia, CKD, GERD, and a prior history of UTI with *E.coli* sensitive to nitrofurantoin, sensitive to TMP/SMX, resistant to cefixime, and resistant to ciprofloxacin. She does not have a catheter placed. She is currently taking donepezil 10 mg PO daily, ramipril 5 mg PO daily, and pantoprazole 40 mg PO daily, and she received levofloxacin 500 mg PO Q24H for 3 days one week ago. CR has no allergies to medications. She lives in a nursing home. Today her BP is 86/42 mmHg, HR 118-142 bpm, RR 35 bpm, SpO₂ 94% on 2L NP, and her weight is 50.8 kg.



CASE 1

CR is transferred to the emergency department of her local hospital. After being assessed by the ED physician, she has 2 peripheral lines placed (18G antecubital and a left femoral) and 2L of NS infused. Owing to lack of response, she is started on norepinephrine 0.08 mcg/kg/min to maintain a MAP of 69. She is intubated using rapid sequence intubation, is initially on an FiO₂ of 0.4 on pressure control ventilation and maintaining oxygen saturations of 95%.

Initial laboratory work was sent, a chest x-ray ordered.

Physical exam revealed no crackles and good breath sounds bilaterally. No signs of meningitis are present, no stigmata of endocarditis noted. Her skin is warm but mottled without signs of rash or other noted abnormalities. She has no sign of ulcerations or bedsores.

A bedside ultrasound revealed significant L sided hydronephrosis. Bladder ultrasound revealed minimal retained urine.

Blood cultures were taken from each of the peripheral lines.

The treating ED physician comes to ask you a few questions about the care of this critically ill patient prior to her transfer to the ICU.

CASE 1 - QUESTIONS

1. What clinical syndrome does CR have?
 - a) MODS
 - b) Sepsis
 - c) Severe Sepsis
 - d) Septic Shock
2. What antibiotic regimen is appropriate for CR at this time?
 - a) ciprofloxacin
 - b) ceftriaxone
 - c) ertapenem
 - d) TMP/SMX
3. What is the preferable fluid management for CR?
 - a) Normal Saline
 - b) Lactated ringers with 40mEq potassium
 - c) Albumin 5%
 - d) Dextran 10%
4. Despite maximal norepinephrine support, CR's BP begins to fall. An arterial blood gas reveals a lactate of 5.3 mmol/L. Her IVC appears dilated on POCUS. What is the best course of action?
 - a) Initiate epinephrine infusion
 - b) Initiate dobutamine
 - c) Initiate vasopressin
 - d) Initiate hydrocortisone

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CHANGE LOG

March 2025

- Spelling and grammatical edits throughout
- Slide 3: Updated legend

July 2025

- Slide 4: Removed SHARE ML as inaccurate
- Slide 15: Chlorhexidine removed from ICU care
- Minor edits to case

October 2025

- No content changes made

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 5: Added FYI